Jennifer A. Kaphammer

M.S. Student, Computer Science
University of Colorado Boulder

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Education

University of Colorado Boulder, CO *August 2026 - Current*
Masters of Science, Computer Science Candidate -
Thesis-Based

University of Colorado Boulder, CO *August 2021 - December 2025*
Bachelor of Arts, Computer Science
GPA: 3.73; Graduating with cum laude Latin honors
Undergraduate Research Opportunities Program (UROP)
Scholar
University of Colorado Boulder, CO *August 2021 - December 2025*
Bachelor of Arts, Music Technology & Media Studies
GPA: 3.73; Graduating with cum laude Latin honors

Research Experience

University of Colorado, Boulder - Peleg Lab - Biofrontiers Institute *June 2024 - May 2025*
Undergraduate Research Assistant
Advisors: Dr. Orit Peleg, Dr. Danielle Chase
University of Colorado, Boulder - Office of Information Technology *Jan 2023 - May 2024*
Buff Portal; Design for America - Student Team Lead
Advisor: Courtney Fell

Scholarships & Grants

Undergraduate Research Opportunities Program (UROP) *Summer 2024*
Research Grant
Provided for summer research in the Peleg Lab under direction of Dr. Orit Peleg & Dr. Danielle Chase

Silver & Gold Scholarship *August 2021 - May 2025*
Tuition Scholarship
Provided for excelling in academics and musical achievements at the University of Colorado, Boulder
Featured on CU Website

CU Esteemed Scholars - Baker Scholarship *August 2021 - May 2025*
Tuition Scholarship
Provided to incoming undergraduate Colorado residents excelling in academics and extraordinary high school community achievement.

Jilling & Ellins Music Scholarship *August 2021 - May 2022*
Tuition Scholarship
Provided for excelling in academics University of Colorado, Boulder.

2021 Be Boulder Scholarship *August 2021 - May 2022*
Tuition Scholarship
Provided for excelling in academics University of Colorado, Boulder

Research Projects

Density & Occupied Fractions of a Honeybee Swarm *June 2024 - May 2025*
Advisor: Dr. Danielle Chase
Conducted data collection on honey bee swarms exploring adaptation to environmental perturbations.
Performed image processing and quantitative analysis using MATLAB to evaluate swarm dynamics.
Applied statistical methods to assess behavioral patterns and correlations within honey bee colonies.
Utilized machine learning and computer vision techniques (PyTorch custom U-Nets, YOLO11) using Python to track and predict bee trajectories.
Collaborated closely with lab members to refine analysis techniques and ensure accurate data interpretation.

Predicting Zebrafish Phenotype Expression from Gene Mutations with a Supervised Machine Learning Model *April 2025 - May 2025*
Utilized Zebrafish Information Network (ZFIN) database to obtain phenotypic and genotypic data on *Danio rerio* in the form of txt files.
Parsed data into csv files to load into pandas DataFrames, utilizing techniques such as one-hot encoding to make large feature matrices for hundreds of thousands of entries.
Applied exploratory K-means clustering where each entry was a fish, and features were binary encoded phenotype alterations. Utilized PCA to see visual fish-cluster compressions.
Developed a MLE logistic regression model with abnormal phenotype being marked as 1-class, normal as 0-class.
Logistic regression model came within 0.98 accuracy / precision, and 0.99 recall.

Noise Pollution in London from a Simulated Pedestrian Perspective *May 2024 - July 2024*
Advisors: Dr. Amy Voida, Dr. Stephen Voida
Joint research project with an undergraduate peer on decibel noise ratings in London.
Collected decibel rating data through different points in Central London and surrounding boroughs using a hand-held decibel meter.
Developed prototypes on noise pollution solutions in heavily-populated areas and theoretical data representations.

Professional Experience

Systems Engineer - Spectrum *February 2025 - August 2026*
Supported and improved reliability of distributed systems running on 20+ Linux VMs, troubleshooting production issues and optimizing system performance.
Configured and enhanced logging/monitoring pipelines using Splunk to support debugging and performance analysis of backend services.
Automated infrastructure and operational workflows using Bash scripting, reducing manual intervention and improving system efficiency.
Managed Kubernetes (Rancher) environments supporting containerized applications, contributing to scalable service deployment.
Implemented network and firewall configurations to support secure service communication across distributed systems.

Systems Engineer Intern - Spectrum *May 2025 - August 2025*
Developed a dynamic health check and failover system to redirect traffic across data centers, ensuring high availability.
Built RESTful APIs in Maven-based Java Spring Boot integrating with Oracle DB, Kubernetes (Rancher), and a GSLB.
Automated health check/failover processes using shell scripts and config maps in Rancher.
Deployed containerized application from dev to production environments with monitoring via Splunk and internal logging.
Authored technical and user documentation with architecture diagrams for scalability.
Collaborated using GitLab/Sourcetree for version control and Jira for issue tracking.

Conference Abstracts

Adaptations and Mechanical Memory in Honeybee Swarms in Response to Temperature Changes *2025*
Chase, D. L., van Hoorn, N., Swanson, A., Kaphammer, J., Moran, D., Reyes, C., Shishkov, O., Lemos, P. A., & Peleg, O.
APS Global Physics Summit 2025

Programming Projects

Astro Adventures *Full-Stack Development*
Technologies: HTML, CSS, JS, Bootstrap, EJS, postgreSQL, Node.js, Docker
Developed a full-stack web app using NASA's REST APIs to provide real-time educational space content.
Conducted unit tests utilizing Mocha to ensure site functionality.
Integrated dynamic updates and a user-friendly interface with a Dockerized JS-based stack.
Managed user accounts with postgreSQL to enable personalized experiences.
Ensured scalability and responsiveness by following Agile practices and using Git for version control.

Live Sound Generation *Interactive Media*
Technologies: JavaScript, Freesoud.org API, Web Audio API
Developed a dynamic sound generator that utilizes the Freesound.org and Web Audio APIs to randomly generate and play sounds for live audiovisual performances.
Implemented both server-side and client-side JavaScript, allowing for seamless integration of sound generation with real-time audiovisual content.
Designed the project to be scalable, with plans for enhanced interactivity on the client side during future live performances.

Technical Skills

Programming Languages: Python, Java, Bash, C++, C, SQL, MATLAB, Supercollider
Frameworks/Libraries: Spring Boot, PyTorch, Numpy, YOLO, Scikit-learn, Matplotlib
Database Management: Oracle SQL Developer, MySQL, PostgreSQL
Technologies/Methods: Maven, Splunk, Rancher, Kubernetes, Jira, ETL, REST, EJS, Shell Scripting, Linux, Docker, Mocha, Object-Oriented Programming, GitHub/Git VCS, Agile Methodology, Human-Centered Design Process, Microsoft Suite, Excel
Lab Processes: Image Analysis, Data Visualization, Radiation safety certification, MATLAB scripted CT scan operation, stinging insect lab handling procedure